SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	D. Sri Anjaneya	**Reg. No.:**	192425219
Section / Batch:	AI & ML / 2024	**Date:**	01/08/2026

Code of Conduct

I D. Sri Anjaneya / 192425219 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _D. Sri Anjaneya_

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4

Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as “govern”, “government”, and “governance”. The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

4.A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.
Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.
Analyze how prefix addition and suffix transformation affect meaning and word class.
Apply the process to decompose each word and map them to a unified base representation. Derive the morphological structure and normalized root for each input word.

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.
Identify suffix patterns and map all variations to a common base form without altering meaning.
Apply the process to extract root forms and classify each transformation. Determine the morphological breakdown and normalized output for each word.

Q1. Morphological Processing of analyzing, analysis, and analytical for Search Engine Indexing

Scenario Understanding:

A search engine receives different forms of the same concept. If these words are treated as separate terms, retrieval accuracy decreases because related documents are not grouped together. A morphological processing system is required to identify the common meaning and store them under a unified representation.

Morphological Processing Approach:

The indexing system follows these steps:

1. **Token Identification**
    - Detect the input words from user queries and documents.
2. **Morphological Decomposition**
    - Split each word into root/base form and affixes.
    - Identify suffixes that create new grammatical forms.
3. **Morphological Classification**
    - Determine whether the change is:
        - **Inflectional** → changes grammatical form without changing meaning.
        - **Derivational** → creates a new word with a new meaning or word class.
4. **Normalization**
    - Convert related word forms into a common root representation.

Morphological Analysis:

Word	Decomposition	Root	Affixes	Type	Explanation
analyzing	analyze + ing	analyze	-ing	Inflectional	Indicates ongoing action (present participle)
analysis	analyze + sis	analyze	-sis	Derivational	Converts action into a concept/noun form
analytical	analyze + ical	analyze	-ical	Derivational	Converts concept into descriptive adjective

Decision and Normalized Output

The system maps all three words to:

Normalized Root = ANALYZE

This improves:

- Search accuracy
- Document retrieval
- Keyword matching

Q2. Morphological Parsing of disagree, agreement, and agreeable for Sentiment Analysis

Scenario Understanding:

Sentiment analysis systems process customer reviews where the same idea may appear in different grammatical forms. The system must understand that disagree, agreement, and agreeable are related but have different meanings.

Morphological Parsing Strategy:

The system performs:

1. **Prefix Detection**
    - Identifies prefixes added before the root.
    - Example: dis- changes the meaning of agree.
2. **Suffix Identification**
    - Detects suffixes that modify word class.
    - Example: -ment changes verb into noun.
3. **Semantic Interpretation**
    - Understands how affixes affect the meaning.
4. **Normalization**
    - Maps words to the common base form.

Morphological Analysis:

Word	Decomposition	Root	Affixes	Type	Effect on Meaning
disagree	dis + agree	agree	Prefix: dis-	Derivational	Creates opposite meaning
agreement	agree + ment	agree	Suffix: -ment	Derivational	Shows the state/result of agreeing
agreeable	agree + able	agree	Suffix: -able	Derivational	Represents a quality or possibility

Decision and Normalized Output:

Normalized Root = AGREE

The system understands:

- disagree → negative sentiment
- agreement → positive relation
- agreeable → positive quality

This improves sentiment classification accuracy.

Q3. Morphology-Based Normalization of govern, government, and governance

Scenario Understanding:

A news aggregation system processes many articles where the same concept appears in different word forms. Words like govern, government, and governance have related meanings but different structures. To improve topic modeling and clustering, the system must identify their common root and normalize them into a single representation.

Morphology-Based Normalization Approach:

The system performs the following steps:

1. **Word Decomposition**
    - Each word is separated into its root form and affixes.
2. **Affix Identification**
    - Suffixes are detected and analyzed to understand how they modify the original word.
3. **Derivational Analysis**
    - The system identifies whether suffixes create new words or change the word class.
4. **Normalization**
    - Different forms are converted into the common root word for better document grouping.

Morphological Analysis:

Word	Breakdown	Root	Affixes	Type	Explanation
govern	govern	govern	None	Base form	Original verb representing the action of controlling or managing
government	govern + ment	govern	-ment	Derivational	Suffix changes the verb into a noun representing an organization or system
governance	govern + ance	govern	-ance	Derivational	Suffix changes the verb into a noun representing the process of governing

Normalized Representation:

Input Word	Normalized Form
govern	govern
government	govern
governance	govern

Decision:

All three words are mapped to the common root **govern**. This helps the text analytics system group related news articles, improve topic modeling accuracy, and reduce vocabulary variations during clustering.

Final Output:

govern, government, and governance → govern

Q4. Morphological Analysis of activate, activation, and reactivation

Scenario Understanding:

A document classification system processes words with different forms but related meanings. Words such as activate, activation, and reactivation represent the same basic concept but contain different prefixes and suffixes. The system needs to identify the root word, analyze derivational changes, and normalize these words for better semantic grouping and indexing.

Morphological Parsing Strategy:

The system follows these steps:

1. **Root Identification**
    - Extract the basic root that carries the main meaning.
2. **Affix Detection**
    - Identify prefixes and suffixes added to the root.
3. **Derivational Analysis**
    - Determine how affixes change the word class and meaning.
4. **Normalization**
    - Map all related forms to a common root representation.

Morphological Analysis:

Word	Breakdown	Root	Affixes	Type	Explanation
activate	act + ivate	act	-ivate	Derivational	Forms a verb representing the action of making something active
activation	activate + ion	act	-ion	Derivational	Converts the verb into a noun representing the process of activating
reactivation	re + activate + ion	act	re-, -ion	Derivational	Prefix **re-** indicates repetition, and suffix **-ion** changes the verb into a noun

Derivational Effects:

- **activate** → refers to performing an action.
- **activation** → refers to the process or result of activating.
- **reactivation** → refers to activating something again.

The prefix **re-** adds the meaning of repetition, while the suffix **-ion** changes the word from a verb form to a noun form.

Normalized Representation:

Input Word	Normalized Form
activate	act
activation	act
reactivation	act

Decision

All three words are mapped to the common root **act**. This normalization helps the document classification system identify related documents even when different derivational forms are used.

Final Output:

activate, activation, and reactivation → act

Q5. Inflectional Normalization of create, creates, and creating

Scenario Understanding:

A search optimization module processes different grammatical forms of the same verb. Words like create, creates, and creating have the same core meaning but differ due to tense and aspect changes. The system must identify these variations and normalize them into a common base form to improve search retrieval and indexing.

Morphology-Based Normalization Approach:

The system follows these steps:

1. **Identify Base Form**
    - Extract the original verb that carries the main meaning.
2. **Detect Inflectional Suffixes**
    - Identify suffixes added for grammatical purposes such as tense or aspect.
3. **Classify Transformations**
    - Determine whether the change represents present tense, third-person form, or continuous action.
4. **Normalize Word Forms**
    - Remove grammatical suffixes and convert all variations into the root form.

Morphological Analysis:

Word	Breakdown	Root Form	Suffix	Type	Explanation
create	create	create	None	Base form	Original verb representing the action of making something
creates	create + s	create	-s	Inflectional	Indicates third-person singular present tense
creating	create + ing	create	-ing	Inflectional	Indicates continuous or ongoing action

Inflectional Transformation Analysis:

- **create** → basic verb form.
- **creates** → suffix **-s** shows subject-based tense variation without changing the meaning.
- **creating** → suffix **-ing** indicates an ongoing action or progressive aspect.

These suffixes only provide grammatical information and do not create a new word or change the original concept.

Normalized Representation:

Input Word	Normalized Form
create	create
creates	create
creating	create

Decision:

All forms are mapped to the common root **create** because the differences are only inflectional changes. This normalization improves search accuracy by allowing the system to retrieve documents containing any grammatical variation of the word.

Final Output:

create, creates, and creating → create

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____

Date: _____	Date: _____	Date: _____